LESSON 8.4

# Write and Represent Inequalities

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can write inequalities to represent real-world situations.

**WHY THIS MATTERS** Detective Navarro is narrowing down suspects. A witness says the suspect is at least 18 years old. The detective writes the inequality a ≥ 18, where a is the suspect's age. How does this help eliminate suspects?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Inequality** *Spanish: Desigualdad* | A math sentence that compares two sides with <, >, ≤, or ≥. | x ≥ 18 |
| **Greater than** *Spanish: Mayor que* | The > sign, showing one number is bigger. | 8 > 3 — eight is larger than three |
| **Less than** *Spanish: Menor que* | The < sign, showing one number is smaller. | 2 < 7 — two is smaller than seven |
| **At least / At most** *Spanish: Al menos / A lo más* | 'At least' means ≥. 'At most' means ≤. | 'At least 10 players' → p ≥ 10 (could be 10, 11, 12, ...); 'At most 5 tries' → t ≤ 5 (could be 5, 4, 3, ...) |
| **Variable** *Spanish: Variable* | A letter that stands for a number that is unknown or can change. | the x in x ≥ 18 |
| **No more than** *Spanish: No más de* | A phrase meaning less than or equal to (≤). | “No more than 8 people” means people ≤ 8. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**A van can carry at most 8 riders. Write an inequality for r, the riders.**

1. Key words: “at most.” Is 8 allowed? Yes.
2. “At most” means less than or equal to: r ≤ 8.
3. Test: 8 works, 5 works, 9 does not.

**Answer:** r ≤ 8

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Clue: 'A bag holds fewer than 10 items.' What does 'fewer than' mean?**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The phrase _____ tells me the symbol is _____ .
- My inequality is _____ because _____ .
- The value _____ (is / is not) a solution because _____ .
- This inequality has many solutions because _____ .

**Word bank** inequality | at least | at most | more than | less than | greater than | symbol | constraint | variable | solution | boundary | range

WATCH OUT!

## A common mistake

A common mistake in Write Inequalities is using a strict symbol (> or <) for a phrase that actually includes the boundary value. For 'the suspect is at least 18 years old,' students often write a > 18 instead of the correct a ≥ 18

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**Level 0** · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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**1** **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The temperature is less than 40 degrees'?

(A) t < 40 (B) t > 40

(C) t ≤ 40 (D) t = 40

STEPS TO TRY

1. Find the comparison phrase: 'less than 40 degrees'. Does 40 itself count, or only temperatures below it?
2. 'Less than' with no 'or equal to' means a strict symbol — no bar underneath — pointing toward the smaller side.
3. Test a value: is 35 degrees 'less than 40'? Yes — so pick the strict symbol that puts t on the smaller side of 40.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'You must be at least 48 inches tall to ride'?

(A) $h \geq 48$

(B) $h > 48$

(C) $h < 48$

(D) $h \leq 48$

**STEPS TO TRY**

1. 'At least 48' — ask yourself: can a rider who is EXACTLY 48 inches tall ride? That decides whether 48 is included.
2. When the boundary value is allowed, you need a symbol with the 'or equal to' bar under it.
3. Allowed heights are 48, 49, 50, and up. Pick the symbol that includes 48 itself AND points toward bigger numbers.

**SHOW YOUR WORK**

**3** **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'A store can hold no more than 75 people'?

(A) $p \leq 75$

(B) $p < 75$

(C) $p \geq 75$

(D) $p > 75$

**STEPS TO TRY**

1. 'No more than 75' — could the store hold exactly 75 people? Could it hold 76?
2. 'No more than' allows the boundary but nothing above it — that calls for a symbol with the 'or equal' bar.
3. Test values: 75 people is fine, 76 is not. Pick the symbol that keeps p at 75 or below.

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The speed limit is more than 25 mph'?

(A) $s > 25$

(B) $s \geq 25$

(C) $s < 25$

(D) $s = 25$

**STEPS TO TRY**

1. 'More than 25' — is exactly 25 included, or only speeds above 25?
2. 'More than' (not 'at least') is strict — no 'or equal' bar under the symbol.
3. Test: 26 works, but 25 itself does not. Pick the strict symbol that puts s above 25.

**SHOW YOUR WORK**

**EXPLAIN YOUR THINKING**

**How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?**

**SENTENCE STARTER** 'At least' uses ≥ instead of > because it means the value ___ be equal to the number, while 'more than' means it must be ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help